Operating Systems · 21CSC202J

Unit 1 · Session 8 · SLO 1 — Solved Worksheet

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Unit â€“ 1 : Session â€“ 8 : SLO â€“ 1 | Kernel Data Structures â€” Symbol Code Puzzle

I. Secret Message

• Decoding the symbols with the key at the bottom of the page reveals:
• "KERNEL DATA STRUCTURES KEEP TRACK OF PROCESSES, MEMORY, FILES AND DEVICES."

II. What Are Kernel Data Structures?

• PROCESS LIST / PCB TABLE â€” one Process Control Block per process (state, registers, PC).
• READY QUEUE and WAIT (device) QUEUES â€— linked lists of process descriptors used by schedulers.
• MEMORY MAPS / PAGE TABLES â€— track free and allocated frames and address translation.
• OPEN FILE TABLE and INODE structures â€— track files opened by processes.
• DEVICE TABLE and I/O request queues â€— track devices and pending operations.